Saveetha Institute of Medical And Technical Sciences
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Queue Using Array

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INTRODUCTION

What is Queue?

- A **Queue** is a linear data structure.
- It follows the **FIFO (First In, First Out)** principle.
- The first element inserted is the first element removed.
- Queue can be implemented using an array or linked list.
- Unlike a stack, where insertion and deletion occur from the same end, a queue performs insertion at the **rear** and deletion from the **front**.

Queue Representation Using Array

- Index

```
0     1     2     3     4
+-----+-----+-----+-----+-----+
| 10  | 20  | 30  |     |     |
+-----+-----+-----+-----+-----+
```

- Front = 0
- Rear = 2

Explanation

- Array stores all queue elements.
- **Front** points to the first element.
- **Rear** points to the last inserted element.
- Maximum number of elements depends on the array size.

Queue Operations

- **Enqueue** – Insert an element at the rear.
- **Dequeue** – Delete an element from the front.
- **Display** – Show all queue elements.
- **Peek** – Display the front element.

1. In a deque (double-ended queue) with elements [5,10,15,20]if an operation is performed to insert the element '25' at the front and then delete an element from the rear, what will be the rear element of the deque?

A.15 B.20 C.25 D.10

2. A linear queue has size 5. The queue contains: [_, _, 30, 40, 50] where front = 2 and rear = 4.What happens when Enqueue(60) is performed?

A.60 is inserted at index 0

B.60 is inserted at index 1

C.Queue Overflow

D.60 replaces 30

3. A priority queue Q is used to implement a stack that stores characters. PUSH (C) is implemented as INSERT (Q,C,K) where K is an appropriate integer key chosen by the implementation. POP is implemented as DELETEMIN (Q). For a sequence of operations, the keys chosen are in

A.non-increasing order
B.non-decreasing order
C.strictly increasing order
D.strictly decreasing order

4. Five people P, Q, R, S and T are standing in a queue. R is standing between P and T. P is just behind Q and Q is second in the queue. Who is second last in the queue?

A.T B.S C.R D.P

5.How many stacks are required to implement a queue? Where no other data structure like arrays or linked lists is available.

A.1 B.2 C.3 D.4

THANK YOU..


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